

UNIT 10

SOLUTION, SUSPENSION & COLLOIDS



Solution:

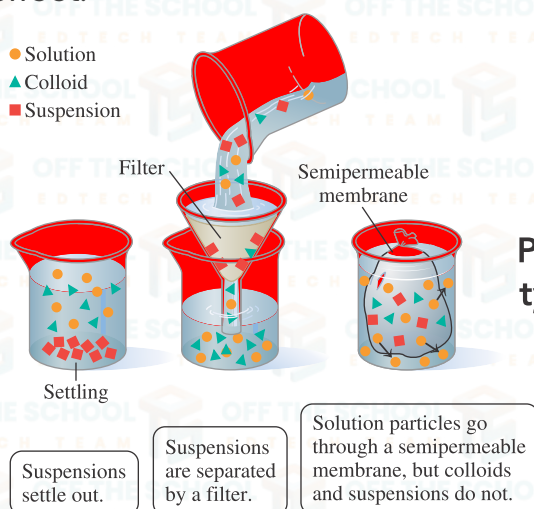
A solution is a homogeneous mixture where solutes are uniformly dissolved at the molecular or ionic level in a solvent, leading to a transparent or translucent mixture that does not settle upon standing.

• Suspension:

A suspension is a heterogeneous mixture with solid particles dispersed in a liquid or gas, which are large enough to settle over time. It exhibits the Tyndall effect and requires agitation to maintain dispersion.

• Colloid:

A colloid is a heterogeneous mixture with tiny particles or droplets evenly dispersed throughout another substance. Colloidal particles are sized between those in solutions and suspensions, do not settle upon standing, and may show the Tyndall effect.



Properties of different types of mixtures.

THE NATURE OF SOLUTION IN LIQUID PHASE:

- **Completely Miscible Liquids:** These are liquids that mix in any amount to form a uniform mixture. For example, methanol and water can mix in any proportion because they have similar types of molecular attractions, like hydrogen bonds for water and methanol, and dispersion forces for benzene and toluene.
- **Partially Miscible Liquids:** These liquids can only mix up to a certain amount. Beyond that, they separate into two layers. For example, if you mix phenol with water, beyond a certain ratio, you'll get a layer with a higher concentration of phenol on top and a layer with more water on the bottom.
- **Incompletely Immiscible Liquids:** These liquids do not mix at all and always form separate layers. An example is water and benzene; no matter how much you mix them; they won't combine into a single uniform liquid.

Solubility:

Solubility is defined as the maximum amount of a solute that can dissolve in a given amount of solvent at a specified temperature and pressure to form a saturated solution. It's often expressed in grams of solute per 100 grams of solvent.

Factors affecting solubility:

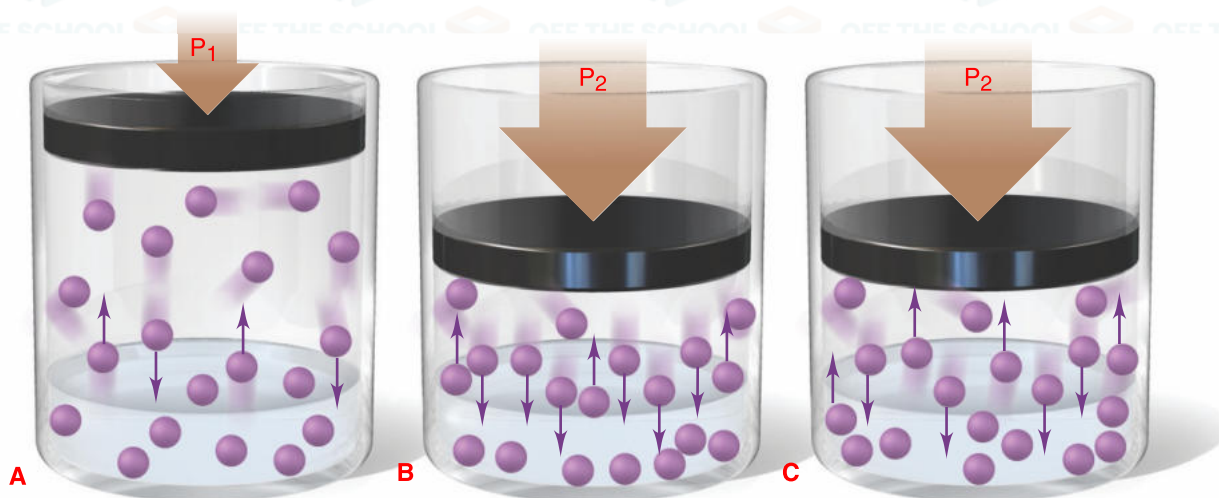
1. **Nature of the Solute and Solvent:** The solubility of a substance depends on the chemical nature of both the solute (the substance being dissolved) and the solvent (the substance the solute is being dissolved into). For example, polar solutes tend to dissolve well in polar solvents (like salt in water), whereas nonpolar solutes dissolve better in nonpolar solvents (like oil in gasoline).

2. Temperature: For most solid solutes, solubility increases with an increase in temperature. However, for gases, solubility typically decreases as the temperature increases.

3. Pressure: The solubility of gases in liquids is directly proportional to the pressure of the gas above the liquid. This relationship is described by Henry's Law. Increasing the pressure increases the solubility of a gas in a liquid.

4. Molecular Size: Larger molecules or ions often have lower solubility than smaller ones due to their difficulty in interacting with the solvent molecules.

5. Polarity: Substances with similar polarities tend to be more soluble in each other (like dissolves like principle). For example, ionic and polar substances are generally more soluble in polar solvents, while nonpolar substances are more soluble in nonpolar solvents.



The effect of pressure on gas solubility.



The effervescence of a soft drink. The bottle was shaken before being opened to dramatize the escape of CO_2 .



Raoult's law:

Introduction: In the study of solutions and their behaviors, Raoult's Law plays a fundamental role. It's named after the French chemist François-Marie Raoult who proposed it in 1887.

Statement of Raoult's Law:

The formal statement of Raoult's Law can be given as follows:

For a solution of non-volatile solute and volatile solvent, the vapor pressure of the solvent above the solution (P_{solution}) is directly proportional to the mole fraction of the solvent (X_{solvent}) in the solution.

The law is mathematically represented as:

$$P = P_{\text{solvent}^\circ} \times X_{\text{solvent}}$$

Here, P represents the vapor pressure of the solution, P_{solvent° is the vapor pressure of the pure solvent, and X_{solvent} is the mole fraction of the solvent in the solution.

When a non-volatile solute is added to the solvent, it occupies some of the surface area, reducing the number of solvent molecules that can escape to the vapor phase. This results in a lower vapor pressure of the solution than that of the pure solvent.

The change in vapor pressure (ΔP) caused by the addition of a non-volatile solute is proportional to the mole fraction of the solute (X_{solute}). This can be represented as:

$$\Delta P = P_{\text{solvent}^\circ} \times X_{\text{solute}}$$

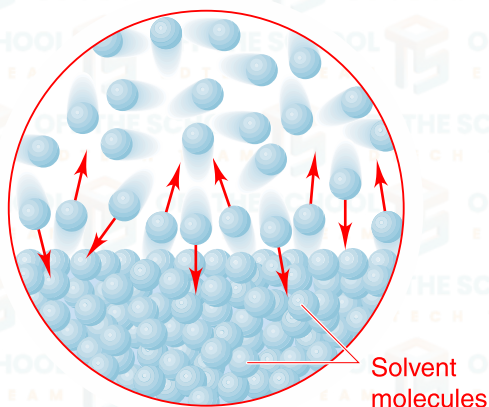
Additionally, the relative lowering of the vapor pressure ($\Delta P/P^\circ$) is equal to the mole fraction of the solute (X_{solute}):

$$\frac{\Delta P}{P^\circ} = X_{\text{solute}}$$

Raoult's Law also extends to solutions where both components are volatile, with each component contributing to the total vapor pressure of the solution in proportion to its mole fraction.

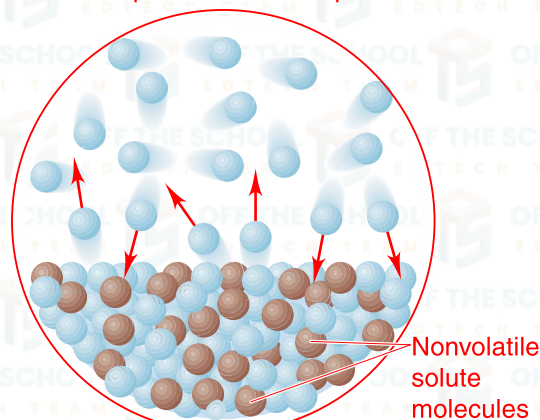
The law is crucial for understanding the properties of solutions and is applied in various scientific and industrial processes.

Equilibrium is reached with a given number of particles in the vapor.



A Pure solvent

Equilibrium is reached with fewer particles in the vapor.



B Solvent and dissolved solute

Effect of solute on the vapor pressure of solution.

A, Equilibrium between a liquid and its vapor occurs when equal numbers of molecules vaporize and condense in a given time.

B, Solute molecules increase the entropy of the solution, so equilibrium occurs at a lower vapor pressure.

Colligative properties of solutions:

Colligative properties of solutions are physical properties that depend on the number of solute particles in a solution rather than the chemical nature of the solute. These properties include:

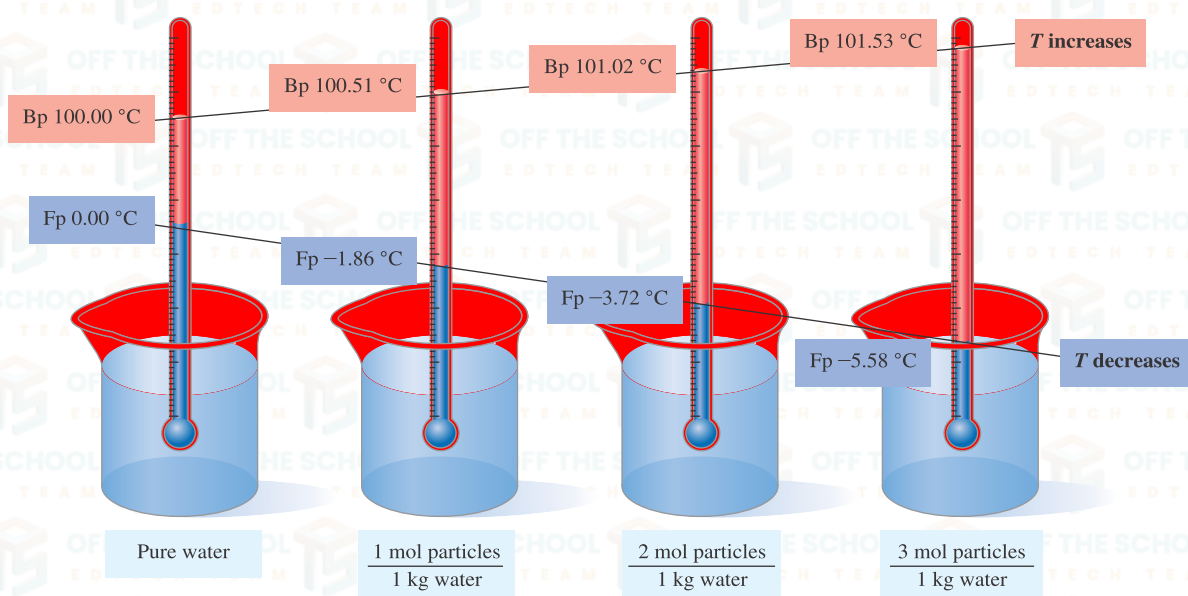
1. Lowering of Vapor Pressure: Adding a non-volatile solute to a solvent reduces the vapor pressure of the solvent. This happens because the solute particles take up space at the surface of the liquid, making fewer solvent molecules available to evaporate. Raoult's Law quantifies this reduction in vapor pressure and states it is directly proportional to the mole fraction of the solute.

2. Elevation of Boiling Point: The boiling point of a solution is higher than that of the pure solvent. This is because a solution with a non-volatile solute has a lower vapor pressure, and it takes more heat to make the vapor pressure of the solution equal to the atmospheric pressure, which is the condition for boiling. This increase in the boiling point is known as the elevation of boiling point.

3. Depression of Freezing Point: The presence of a solute also lowers the freezing point of a solution compared to the pure solvent. At the freezing point, the vapor pressures of the liquid and solid phases of the pure solvent are equal.

When a solute is added, it decreases the vapor pressure of the liquid phase, so the temperature must be lowered further to reach the point where the vapor pressures of the solid and liquid are equal again. This lowered temperature at which the solution freezes is the depression of the freezing point.

The extent to which these properties change depends on the number of solute particles and is irrespective of their identity. This is why colligative properties are valuable for determining molecular weights and for other practical applications, like adding salt to water to cook food faster or to ice to melt it.



As the concentration of a solution increases, the freezing point is lowered and the boiling point is raised.

Osmotic pressure:

Osmotic pressure is a fundamental colligative property of solutions, which depends on the number of solute particles present in the solution and is not dependent on their identity. It's a measure of the tendency of solvent molecules to pass through a semipermeable membrane to equalize solute concentrations on both sides. Osmotic pressure can be thought of as the pressure required to prevent this migration of solvent molecules.

Here are some examples and explanations

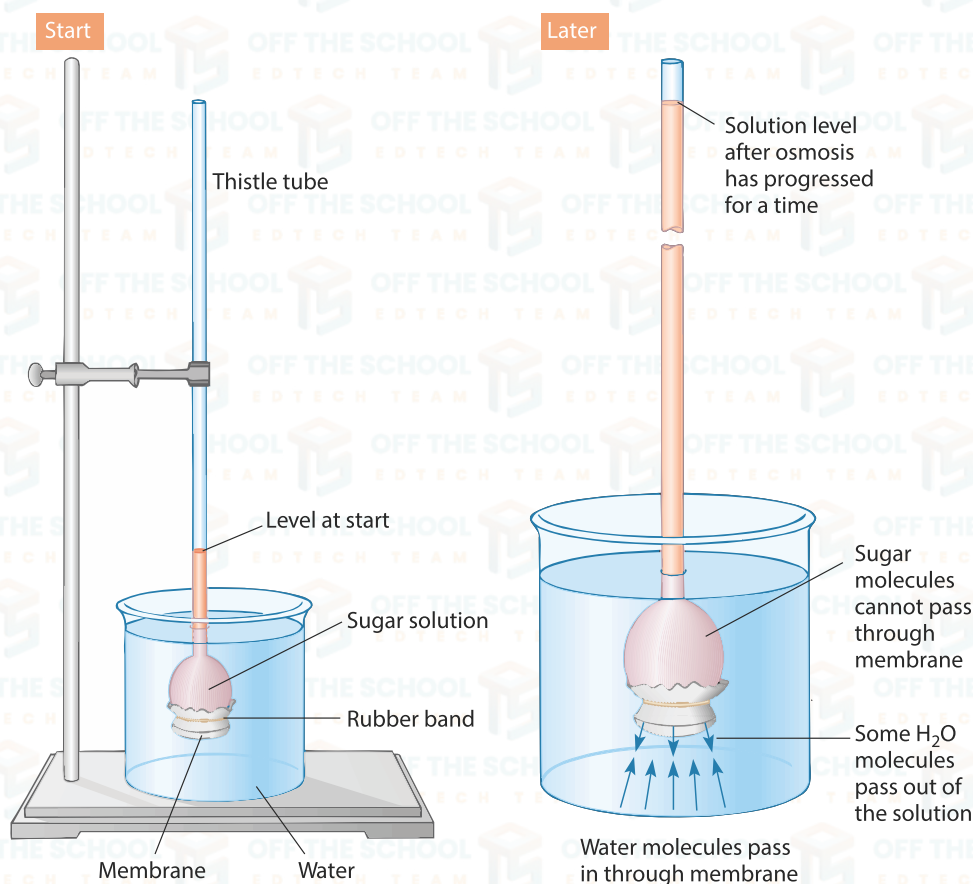
1. Osmosis in Biological Systems: Osmosis is crucial in biological systems where it's responsible for the movement of water and nutrients into cells and the removal of waste products. When two solutions have equal osmotic pressure across a semipermeable membrane, they are called isotonic. In medical treatments, isotonic saline solutions are used to keep cells from shrinking or swelling.

2. Hemolysis: This is a process where red blood cells (RBCs) swell and burst due to an influx of water when placed in a hypotonic solution (a solution with a lower solute concentration than the inside of the cell), which occurs through osmosis.

3. Aquatic Life: Freshwater fish constantly absorb water through their semipermeable skin due to osmosis, as their body fluid is saltier than the freshwater environment they live in.

4. Plant Absorption: Plants absorb water through their entire surface, mainly at the root hairs, by osmosis. This water uptake is essential for the plant's survival, allowing it to obtain necessary nutrients dissolved in the water.

5. Food Preservation: In the food industry, osmosis is exploited for preservation. For instance, jam is made with high amounts of sugar, and pickles with salt, creating a hypertonic environment that draws water out of microbial cells via



osmosis, inhibiting their growth and prolonging the shelf life of the product.

6. Digestion: After food is digested in the stomach and intestines, the resulting nutrients are absorbed into the body tissues through the semipermeable lining of the small intestine by osmosis.

Osmotic pressure is represented by the Greek letter π and is typically measured in units of pressure such as atmospheres. It's important not only in chemistry and biology but also in medicine and food science, where osmotic pressure must be carefully controlled to maintain health and stability in living organisms and food products.

Colloids:

These are mixtures where very small particles of one substance are distributed evenly throughout another substance. These particles are larger than molecules but too small to settle out or be filtered easily, and they don't dissolve in the dispersion medium (the substance in which they are distributed).

Types of Colloids:

Colloids can be classified based on the states of the dispersion medium and the dispersed phase. Here are some types:

1. Aerosols: Dispersed liquid or solid particles in a gas (e.g., fog, spray, and smoke).

2. Foams: Gas particles dispersed in a liquid (e.g., shaving cream) or a solid (e.g., foam rubber).

3. Emulsions: Liquid droplets dispersed in another liquid (e.g., milk or mayonnaise).

4. Sols: Solid particles in a liquid (e.g., paint or blood).

5. Gels: Liquid in a solid, which forms a jelly-like structure (e.g., gelatin or jelly).

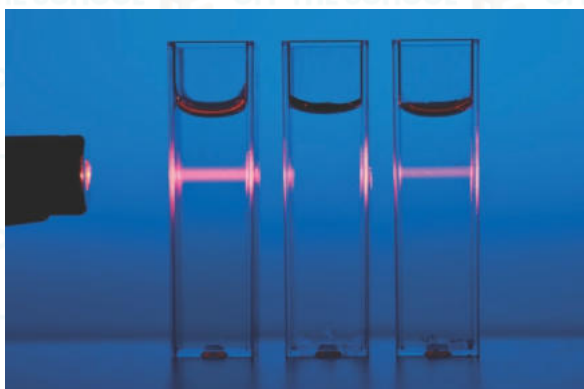
6. Solid Sols: Solid particles in a solid (e.g., colored glass or certain alloys).

7. Solid Foams: Gas in a solid (e.g., pumice or Styrofoam).

Properties of Colloids:

1. Heterogeneous Nature: Colloids consist of two distinct phases—the dispersed phase and the dispersion medium—yet appear homogeneous to the naked eye.

2. Tyndall Effect: Colloidal particles are large enough to scatter light, which makes a beam of light visible as it passes through the colloid.



The Tyndall effect. Light is scattered by colloidal particles (left and right) but not by dissolved particles (center).



A familiar example of the Tyndall effect: headlights illuminating fog.

3. Brownian Motion: The particles in a colloid move in a random, zigzag fashion, which helps keep them dispersed and suspended.

4. Stability: Colloidal particles usually do not settle out on standing and cannot be separated by standard filtering techniques.

5. Charged Particles: Colloidal particles often carry an electric charge, which keeps them from clumping together as the like charges repel each other, maintaining the dispersion.



6. Viscosity: Colloidal mixtures can be quite thick and viscous, more so than the dispersion medium by itself.

Colloids play a vital role in our everyday lives, from the foods we eat to the products we use, and even in biological systems, where they facilitate many of the body's functions.

Molarity:

It is a measure of the concentration of a solute in a solution and is defined as the number of moles of solute per liter of solution. Its formula is:

Molality:

It is defined as the number of moles of solute per kilogram of solvent.

How The Vapour Pressure Of A Solution Is Lowered By Adding Non-Volatile Solute:

In a solution, when a non-volatile solute is added to a solvent, the vapor pressure is lowered. This happens because the solute particles occupy space at the surface of the solution, which means there are fewer solvent particles at the surface that can escape into the vapor phase. With fewer solvent particles available to evaporate, the vapor pressure above the solution decreases.

This effect is explained by Raoult's Law, which states that the vapor pressure of an ideal solution is directly proportional to the mole fraction of the solvent in the solution. Adding a non-volatile solute reduces the mole fraction of the solvent, thus lowering the vapor pressure. (Q no.03 of STBB, Chemistry page #217)

Reason:

The solubility of gases in liquids, such as oxygen in water, is governed by Henry's Law, which states that the solubility of a gas in a liquid is directly proportional to the pressure of that gas above the liquid. Therefore, as pressure increases, more gas molecules are "pushed" into the solution.

On the other hand, the solubility of solids in liquids, like glucose in water, is not significantly affected by pressure. This is because solids and liquids are much less compressible than gases; changes in pressure do not greatly alter their volume or solubility. Instead, the solubility of solids in liquids is more strongly influenced by temperature. (Q no.05 of STBB, Chemistry page #217)

